

“Briefly Entertaining but Pointless”: Perceived Benefits & Risks of Social Robots in the Home

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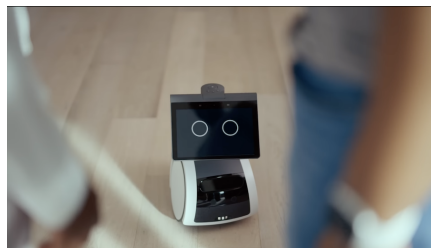
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(a) Jibo



(b) Astro



(c) Alpha 2

Figure 1: The three robot assistants as shown to participants. Pictures taken from the presented videos.

ABSTRACT

In contrast to the adoption of personal assistants, social robots have yet to break into the domestic market. Several manufacturers have introduced and quickly retracted their social robots for the home. We report on a survey study ($N = 50$) to understand potential users’ perceptions towards these social robots. Participants were presented with videos of three domestic social robots and subsequently provided their perception of these in terms of perceived benefits, attraction, privacy risk, usage intention, and capabilities. While participants perceived hedonic and utilitarian benefits, we found a low intention of future adoption of these devices. Further, our findings showed that owners of personal assistants perceived significantly higher hedonic benefits, fewer privacy risks, and higher intention to use domestic social robots. Our work provides an initial step towards understanding perceptions towards social robots and how previous exposure to domestic AI shapes users’ perceptions.

CCS CONCEPTS

• **Human-centered computing** → *Personal digital assistants*; **Empirical studies in HCI**; **User studies**.

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KEYWORDS

Social robots, personal assistants, robot assistants, domestic robots

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1 INTRODUCTION

Following the introduction of Siri in 2011, the number of people with access to personal voice assistants has rapidly increased. Personal assistants can be used for a wide range of tasks, such as setting alarms or sending messages. As of 2022, 39% of private households in the UK own at least one digital personal assistant (PA) or smart speaker [26]. The domestic context in which these devices are used is characterised by a high variation in terms of devices and services that form a household’s digital infrastructure. These include products such as Alexa or Google Home, service robots such as robot vacuum cleaners [23], as well as other robots aimed at domestic use (e.g., Jibo [7] or Amazon Astro [1]). The latter device type are often referred to as ‘social robots’, and promises to bring additional functionality or entertainment to the home through expressive movement and locomotion.

Over the last few years, a multitude of social robots that use movement and locomotion have entered the consumer market. Examples include Jibo, which, through movement, promises the ability to ‘express a wide range of human-like emotions’ [7]. However,

Jibo was discontinued within its first year after release¹. Another example is the Kuri robot [22], a social robot for the home that promises to use locomotion to present ‘fluid and natural’ movement. Within two years of its release, Mayfield Robotics ceased production. These examples indicate a lack of consumer interest or need for this type of device, possibly due to a technology-centred rather than user-centred strategy towards introducing social robots for the home.

In contrast to the challenges faced by social robots trying to enter the market—framed as mobile personal assistants (i.e., Robot Assistants)—personal assistants have found large uptake in the domestic consumer market. As such, it remains to be seen to what extent the additional functionality promised by social robots—such as their ability to move or utilise facial expressions—is perceived as desirable by potential end-users. Furthermore, as new social robots keep appearing on the market (e.g., Samsung Ballie² or the Misa Robot³), the HCI literature does not yet provide a clear understanding about what the deciding factors are for device use and non-use, and which values users see in these robots. In this paper we seek to answer the following research question: How do a social robot’s features and user characteristics affect user perception of domestic social robots?

We report on an online study investigating characteristics contributing to social robots’ acceptance and desirability, informing the future development of personal assistants in the home. Our participants were presented with three videos of representative social robots targeting the domestic market (as shown in Figure 1). Furthermore, we conducted a comparative study of perceptions towards these new devices between participants with ownership of PAs and those without. Our results show that participants with prior exposure to domestic AI systems (i.e., PAs) perceive significantly higher hedonic benefits, fewer privacy risks, and a higher intention of future adoption of this device category for the home. Furthermore, we identified significant differences in perception towards these devices based on gender and education level.

2 RELATED WORK

We outline contemporary research on social robots in the domestic space and prior work on the challenges of the adoption of social robots by users.

2.1 Social robots in the domestic space

Embodied AI systems, such as robots, are used for a number of different roles in domestic spaces, e.g., private homes or assisted living facilities. Examples include the use of these systems to reduce loneliness [10, 19, 21], provide smart home control [17], facilitate child development [3, 25], or promote physical activation for elderly [4].

Luria et al. [17] conducted a study on the utilitarian qualities of a social robot as a tangible device for smart home control. The design of the robot is inspired by the morphology of a microscope [16], yet uses fluid movement reminding of the movement expressed by Jibo

(see Figure 1a). The authors argue that the use of physical embodiment combined with expressive motion as feedback modality, in contrast to, e.g., voice-based interfaces or variations of touch-based interfaces, can provide the user with a higher sense of situational awareness within the home. Furthermore, the physical presence of the social robot was perceived as the most enjoyable interaction modality (as compared to voice or variations of touch-based interfaces). As such, the smart home robot provided hedonic benefits while fulfilling a specific use case.

To combat loneliness and social isolation amongst single-person households, Jeong et al. [10] developed a social robot which mediates living noise between individual households. Furthermore, Jeong et al. showed that the multi-modal information provided by an expressive social robot creates a feeling of being physically closer to friends and brings a sense of co-residency despite participants living alone, thereby reducing the feeling of loneliness. This demonstrates how the use of bespoke social robots can successfully complete the specific use-case for which it was designed.

2.2 Adoption Challenges of Social Robots

The adoption challenges faced by social robots have been studied in several studies [5, 6, 9, 11, 13]. De Graaf et al. [6] developed the Karotz robot and deployed it in 102 households for up to six months. The authors were interested in the discontinuation of use (i.e., non-use) and the reasons for this. They tracked when people stopped using the robot and obtained insights into why through questionnaires and interviews at regular intervals following the robot’s introduction. Participants who discontinued their use listed ‘Need not satisfied’ (27%) and ‘Replaced by other device’ (27%) as their primary reasons. Furthermore, participants described the robot’s lack of added value as a reason for discontinuing. The authors state that for social robots to be successful, they need to be “[...] enjoyable and easy to use or (socially) predictable to capture users in the short-term, and functionally relevant and possess enhanced social behaviours to keep those users in the longer-term.” Similar observations are presented by Hoffman [9]. The article argues that one of the key problems these devices face is the lack of a key long-term use cases to maintain engagement beyond the novelty phase. Furthermore, it points to a shortage of storytellers involved in robot development, which results in a lack of emotional engagement in providing the robots with the means to create value in the home.

Highly relevant to our work is McLean et al.’s study on variables influencing the adoption and use of voice assistants [18]. McLean et al. propose and evaluate a model for using these devices based on the Uses and Gratification theory [12]. Through an evaluation with over 700 voice assistant users, the authors identified factors that affect the usage of voice assistants. For example, users do not use voice assistants for enjoyment but rather for their usefulness and convenience in completing tasks. We build on McLean et al.’s work and extend it to the domain of social robots.

3 STUDY

We conducted an online survey to identify perceptions towards social robots aimed at the domestic space (e.g., Amazon Astro or Jibo). Our primary focus was to identify the users’ perception of

¹In 2020, Jibo was acquired by NTT Corporation, attempting to bring Jibo to different markets.

²<https://news.samsung.com/us/samsung-ballie-ces-2020/>

³<https://www.hey-misa.com>

social robots' benefits through their unique features that add functionality (e.g., movement, facial expressivity, or locomotion). These features distinguish social robots from existing personal assistants (e.g., Amazon Echo Dot or the Google Home).

Previous research has shown that mere exposure to robots may influence how people perceive them (e.g., [2, 24]). Based on these findings, we hypothesised that owning similar domestic technologies, such as personal assistants, may impact how people perceive domestic social robots. To investigate this, we recruited participants with and without embodied personal assistants.

3.1 Participants

We recruited our participants through the online crowd-sourcing platform Prolific. Following the data collection, we removed participants who failed to answer a simple attention control question (i.e., 'Please select somewhat disagree'). The final data set consisted of responses from 50 participants (27 female, 23 male, average age 38.46, SD: 13.44). We recruited an even number of participants who owned a personal assistant (PA) and those who did not (noPA). The average completion time of the entire study was 14 min 15 sec. Participants were compensated at an hourly rate of £9.00.

3.2 Measurements

To understand user perceptions of domestic social robots, we investigated seven acceptance factors. Specifically, we use the following multi-item scales: Hedonic Benefits (i.e., deriving pleasure), Utilitarian Benefits (i.e., deriving functionality), Symbolic Benefits (i.e., deriving favourable impressions through device usage [8]), Social Attraction, Perceived Privacy Risk, Intention to Use, and Perceived Capabilities.

The first six were identified by McLean and Osei-Frimpong [18] as motivating factors for the adoption of in-home personal assistants. As we were interested in identifying the perceived benefit of devices not yet widely distributed, the individual questions were adjusted in two ways. First, we changed the focus from voice assistants to 'robot assistants'. Second, we rephrased the questions to inquire about imagined future interactions. For example, 'Completing tasks with my voice assistant makes my life easier' [18] was rephrased to 'I believe that completing tasks with a robot assistant will make my life easier'.

Considering that our participants judged the devices based on videos and not actual usage, we added the additional construct of Perceived Capability. The purpose was to assess the extent to which participants believed those devices would function in reality as they did in the promotional videos. We added two additional items to measure Perceived Capabilities ("I believe that the devices will be capable of performing the tasks shown in the videos without problems." and "I believe that robot assistants will be capable of performing a wide variety of tasks without problems."). All questions were rated using a seven-point Likert scale ranging from (1) 'Strongly disagree' to (7) 'Strongly agree'.

3.3 Procedure

Following an introduction of the study's purpose, participants gave informed consent and were redirected to the demographic section of the questionnaire. This was followed by a page providing a short

description of the remainder of the questionnaire, as well as an introduction to the device category of 'Robot Assistants':

'On the following three pages, you will see short videos on how personal assistants could evolve. This type of device is called Robot Assistant. These devices are very similar to existing personal assistants - such as Amazon Echo, Apple HomePod, or Google Home - but come with additional capabilities.'

Subsequently, participants watched three videos of social robots in the role of robot assistants. As these devices are not yet well distributed or even fully commercially available, participants described their perceptions of these devices purely based on the videos⁴. The devices included were Jibo, Astro, and Alpha 2 (see Figure 1). We selected these devices as they present different degrees of anthropomorphism, movement/locomotion capabilities, and social expressivity through facial expressions. Participants watched these videos in randomised order to account for potential ordering effects. All participants watched all three videos. Following the presentation of the videos, participants answered the questionnaire as described in Section 3.2. To provide participants with the opportunity to contextualise their perception of the devices, we concluded the questionnaire with two open questions: (1) 'Please share your opinion on robot assistants as shown in these videos. Feel free to share anything that comes to your mind, be it positive or negative.' and (2) 'How does your opinion of these robot assistants compare to your opinion on voice assistants (e.g., Google Home, Amazon Echo) - please be as detailed as can be in your response.'

4 RESULTS

This section will present both the findings of the questionnaire and contextualise these through the answers to the two open-ended questions.

4.1 Perceptions on Domestic Robot Assistants

Our data analysis aimed not only to reveal general perceptions of robot assistants but also to assess whether those differed significantly among demographic groups.

First, we performed an exploratory data analysis to identify overarching trends regarding motivating factors that could lead to the future adoption of robot assistants (e.g., Hedonic, Utilitarian, and Perceived Privacy Risk). The reliability analysis of the scales used in the study showed high internal consistency, with Cronbach's alpha ratings ranging from .92 to .97. Simple observations of averages (see Table 1) reveal a relatively low Intention to Use ($M = 3.65$, $SD = 1.42$). This rating on a seven-point scale corresponds to "neither agree nor disagree" regarding the intention to use this technology in the future. This could be partially explained by the high score of Perceived Privacy Risk (for mean comparison, see Table 1), which negatively affects the motivation to use these technologies in the domestic context. On the side of positive factors, Hedonic and Utilitarian Benefits received the highest average scores, while Social Attraction and Symbolic Benefits the lowest. Regarding Perceived Capability, our participants rated the robot assistants with a neutral score.

Second, we investigated whether any demographic factors influenced perceptions of Robot Assistants. Our sample regarding

⁴Jibo (151 sec): <https://youtu.be/H0h20jRA5M0>; Alpha2 (144 sec): <https://youtu.be/E1AtfHm4hf8>; Astro (169 sec): <https://youtu.be/sj1t3msy8dc>

	Cronbach's alpha	Mean	St. Dev.
Hedonic Benefits	.97	5.1	1.46
Utilitarian Benefits	.97	4.71	1.61
Symbolic Benefits	.93	3.02	1.46
Social Attraction	.93	3.52	1.44
Perceived Capability	.92	4.23	1.54
Perceived Privacy Risk	.92	5.38	1.48
Intention to Use	.93	3.65	1.42

Table 1: Descriptive statistics showing Cronbach's alpha, average rating, and standard deviation for all study variables.

personal assistant ownership was evenly split ($N = 25$). We performed independent sample t-tests for all acceptance factors with personal assistant ownership as the dependent variable. We found significant differences in perceptions of Hedonic Benefits, Perceived Privacy Risk, and Intention to Use (for details about t statistics, p -values, effect size, and mean scores, see Table 2). Participants who owned personal assistants reported significantly higher Intention to Use a Robot Assistant in the future than those who did not. Owning a personal assistant also resulted in higher Hedonic Benefit perceptions and less Perceived Privacy Risk of domestic social robots. The differences were not statistically significant for the remaining independent variables, even though averages were consistently higher for participants owning personal assistants.

We split our sample into participants with a university degree ($N = 27$) and those without ($N = 23$), leading to two similar-sized groups. Subsequent t-tests showed that education did account for differences in Social Attraction, Perceived Privacy Risk and Perceived Capabilities. Highly educated participants rated the Robot Assistants lower in Social Attraction and Perceived Capabilities while also considering Perceived Privacy Risk to be much higher than those with less education.

Our sample was also relatively balanced regarding gender (27 female, 23 male). Independent-sample t-tests did not reveal any significant difference in acceptance factors apart from Utilitarian Benefits. Female participants perceived higher Utilitarian Benefits in Robot Assistants as compared to males. We found no significant differences based on age.

Finally, for Technology Expertise, we only found significant differences in perceptions of Symbolic Benefits. Participants with high technical expertise rated the Symbolic Benefits of Robot Assistants lower than those with less expertise.

4.2 Open-ended Responses

In addition to the questionnaire, we presented participants with two open-ended questions (see Section 3.3). The qualitative data was coded independently by two researchers. We identified common topics that were discussed, the most prominent of which were related to the Perceived Privacy Risk (mentioned by 19 participants) and Social Robots Utilitarian Benefits (mentioned by 20 participants). Participants' quotes will be represented as (PX_{PA}) and (PX_{NoPA}) depending on whether they own personal assistants.

Participants in both groups expressed concerns about the invasion of privacy and the amount of data these devices will be able to collect.

"I am intrigued by the idea but am somewhat concerned about the use of the data collected by the robot" [P7_{PA}]

The concern for privacy raised by some participants, combined with the lack of utilitarian value that these devices provide, justifies the need for personal data.

"I think they are creepy and intrusive. [...] They must be mining massive amounts of personal data that are useful to their manufacturers while being of little use to their owners." [P16_{NoPA}]

"Briefly entertaining, but pointless and a way to farm personal interactions out to a machine..." [P9_{NoPA}]

Not surprisingly, we also found clear differences in sentiment provided by participants, depending on whether participants already had experience with AI in the home or not (i.e., PA ownership). These differences were also reflected in the quantitative data (see Section 4.1). Participants expressed opinions related to the Intention to Use this type of device. While none of the quotes from the noPA group mentioned willingness to use these devices—they were expressing strong feelings about not wanting to use them, six participants of the PA group expressed interest in using these devices in the future. For example:

"I think they seem great and I would want to use them" [P4_{PA}]

"I think they will be something I'd look at in future. [...] I can see how robots could be very useful [...]" [P14_{PA}]

Interestingly participants in both groups, while not always seeing value in the devices for themselves, identified use cases in which these devices might be of use. Specifically, participants point to the potential usefulness of these devices for people with disabilities:

"They have potential for helping people with disabilities." [P11_{NoPA}]

"I think they will be a great addition to people's lives, especially for people that may be disabled, have ill health or live alone." [P3_{PA}]

5 DISCUSSION

Our study revealed significant differences between perceptions towards social robots, depending on the pre-existing exposure to domestic embodied AI (i.e., personal assistants). Similar findings related to the impact of exposure on positive user perception have been identified for other robots, such as elder care robots [2] and industrial robots [24].

Specifically, we highlight the perceived differences in opinion towards robot assistants on three acceptance factors: Hedonic Benefits, Perceived Privacy Risk, and Intention to Use. While participants in both groups expressed that the devices seem fun to interact with—albeit this was presented to a lesser extent by the noPA group—we identified a significant difference. The higher Hedonic Benefit rating in the PA condition might indicate a higher sense of openness—and less fear or nervousness—towards this new device category due to the prior exposure to PA's providing a different frame of reference from existing domestic devices. The higher degree of experience, leading to less scepticism, is further supported by the significant

			t(48)	p-value	Cohen’s d
PA ownership					
	Yes (<i>N</i> = 25)	No (<i>N</i> = 25)			
Hedonic Benefits	5.51 (1.15)	4.69 (1.64)	-2.03	.048	1.42
Perceived Privacy Risk	4.89 (1.55)	5.88 (1.24)	2.49	.016	1.41
Intention to Use	4.33 (1.16)	2.96 (1.33)	-3.89	.001	1.25
University degree					
	Yes (<i>N</i> = 27)	No (<i>N</i> = 23)			
Social Attraction	3.14 (1.24)	3.97 (1.56)	2.11	.040	1.40
Perceived Privacy Risk	5.91 (1.06)	4.77 (1.67)	-2.91	.005	1.37
Perceived Capability	3.67 (1.43)	4.89 (1.43)	3.02	.004	1.43
Gender					
	Female (<i>N</i> = 27)	Male (<i>N</i> = 23)			
Utilitarian benefits	5.18 (1.25)	4.16 (1.83)	2.26	.024	1.40
Technology expertise					
	High (<i>N</i> = 21)	Low (<i>N</i> = 29)			
Symbolic benefits	2.51 (1.17)	3.40 (1.56)	2.19	.033	1.41

Table 2: Summary of significant differences across demographic variables and acceptance factors. Standard deviations are reported in brackets. The last two columns show the significance level of the t-test and the effect size (Cohen’s d).

differences in Perceived Privacy Risk. As described in Section 4, participants in the noPA group had a significantly higher rating for Perceived Privacy Risk. This might be related to the mobile nature of the robot, which could result in a more intrusive way to collect personal data (e.g., “[...] they can move across flat floors and have the potential to film you as well as record sound.” [P16_{NoPA}]).

Our findings align well with existing work [9], exemplified by P9_{NoPA}, who describes the current state of domestic social robots as illustrated in the videos as “*briefly entertaining, but pointless*”. For these robots to be successfully adopted into the domestic market, they need a use-case that extends beyond ‘*briefly entertaining*’ and provides them with long-term utility surpassing the novelty period (i.e., making them less ‘*pointless*’). As with other devices and services, e.g., Amazon Alexa or the Google search engine, participants have privacy concerns regarding the data they collect and how it is used. Therefore, we argue that for social robots to become successful, use cases need to be identified that provide a clear value outweighing the privacy concerns caused. Furthermore, there needs to be transparency in relation to what the collected data is used for and how this provides additional value for the user. An example of this is service robots in the home (e.g., vacuum cleaning robots). These devices have cameras and locomotion, just as the robot assistant investigated in this paper. Yet, they are widely adopted in households—potentially due to their clear value proposition (i.e., tackling an undesirable task such as cleaning).

For these devices, the user can identify the additional value provided by, e.g., the camera, as it is used for object detection, leading to an immediate improvement of the robot’s service. As multiple participants point out (e.g., P3_{PA}, P11_{NoPA}, P23_{NoPA}) social robots in the home lack a concrete use case. However, a frequent proposal in both groups is the potential usefulness of this device category for people with physical disabilities. This would require that domestic social robots advance to the point at which they can provide a higher degree of autonomy to the user. Pantofaru et al. [20] have investigated potential tasks that are desirable for domestic service robots. In line with their work, future work could identify use cases desirable for social robots in the home. This

could be combined with a confirmation of the utility domestic social robots could provide to people with varying needs, thereby expanding on existing research investigating social companion robots (e.g., [15]).

5.1 Limitations

As we investigated perceptions of devices which are not commercially available or have a very low distribution, we used manufacturer-created marketing videos to showcase different social robots. However, as these were highly polished videos, they did not showcase the interaction failures that can occur (e.g., the device does not understand a command or gets stuck). Future work could expand by conducting an in-the-wild study using social robots to identify the real-world value they can provide. Furthermore, as the video material was in English, we recruited exclusively from the UK/US. However, previous research has shown that perceptions of robots can differ across different cultures [14]. Therefore, the findings presented here are not necessarily applicable to other cultural contexts.

6 CONCLUSION

We investigated perceptions towards social robots in the home through an online survey (*N* = 50) demonstrating three domestic social robots (Jibo, Astro, Alpha 2) to participants with (*N* = 25) and without (*N* = 25) personal assistants. While we found that participants perceived some Hedonic and Utilitarian Benefits of these devices, the overall Intention to Use them was relatively low. Furthermore, our findings show that owners of personal assistants perceived significantly higher Hedonic Benefits, fewer Perceived Privacy Risks, and a higher Intention to Use them in the future. Lastly, we discuss concerns of privacy risk in relation to the absence of a clear utilitarian value to make social robots in the home desirable, as well as the need for future work to identify well-defined use cases. With this work, we provide an initial step towards understanding perceptions towards social robots in the home and

highlight the impact of previous exposure to domestic AI on shaping users' perceptions.

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